

Step 9: Use of Generative AI (Optional)

How Generative AI was used in the project

Throughout my capstone project, I used ChatGPT and Microsoft Copilot as a collaborative AI assistant to support both the technical and research aspects of the study. AI assisted me in generating initial EDA code and deployment preparation. It also helped in debugging code, explain statistical concepts, generating visualizations, draft sections of the report, writing the README and GitHub documentation, and preparing presentation contents.

Beyond programming, I used AI to refine the research methodology, strengthen the literature review, improve academic writing, and ensure consistency across the different chapters of the report. However, all AI-generated outputs were carefully reviewed, validated, tested, and adapted to the specific requirements of the project. The final design decisions, model selection, interpretation of results, and conclusions remained my responsibility.

Prompt Examples:

1. Generating Visualization

Prompt: please make a figure for this: Public Data Sources | ▼ Company Selection | ▼ News Collection | ▼ Article Screening | ▼ Metadata Recording | ▼ Dataset Validation | ▼ Final Sentiment Dataset.

2. Debugging Code

Prompt:

have error when run comparative calibration curve: -----

NameError Traceback (most recent call last)

/tmp/ipykernel_903/3707617261.py in <cell line: 0>()

```
2
3 plt.plot(
----> 4     prob_pred_lr,
5     prob_true_lr,
6     marker="o",
```

NameError: name 'prob_pred_lr' is not defined<Figure size 800x700 with 0 Axes>

3. Generating Initial Code/Updated Code

- a. Prompt: give the code to add after my model evaluation for logistic regression calibration.
- b. Prompt: please help me update this SHAP codes for logistic regression, random forest and xgboost

4. Preparing presentation contents

Prompt:

please generate a technical presentation for my capstone project with audiences are my professor, peers and panel. the presentation should be 10-12 slides only. The following must be included in the presentation:

Step 1: Problem Understanding & Framing

- Frame the business and data science problem clearly.
- Define whether it's a classification, regression, recommendation, anomaly detection, or clustering task. Deliverable: Clear problem statement + task type + target metric.

Step 2: Data Collection & Understanding

- Datasets used.
- Summarize feature types, missing values, outliers, etc.
- Provide a data dictionary (variables, types, units, allowed values).
- Specify success metrics.

Step 3 Data Preprocessing, Applied EDA & Feature Engineering

- Clean data: Handle nulls, duplicates and outliers.
- Engineer features: explain briefly and what were the used like example domain-derived features.
- Applied EDA: Distributions, relationships, etc.
- Feature importance & explainability: SHAP, or model-based importances.
- Feature selection: explain briefly.
- Dimensionality reduction: explain briefly. include justifications.

Step 4: Model Implementation

- Experiment with appropriate models:
 - o Supervised:
 - o Unsupervised:
- Evaluation: give metrics and compare with relevant metrics.
- Reproducibility: explain briefly.

Step 5: Critical Thinking → Ethical AI & Bias Auditing

- Explain model decisions (e.g., SHAP).

- Address limitations (imbalance, leakages, overfitting).
- Bias detection & fairness audits: explain briefly. Use the attached final report for reference. include also conclusion.